

Q1.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

To find Rank we'll convert A into its echlon form.

$$\hookrightarrow R_2 - R_1 \rightarrow R_3 - R_1 \rightarrow R_4 - R_1$$

$$\text{Echlon form of } A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\text{Rank} \Rightarrow \text{no. of non zero rows} \Rightarrow 1}$$

For eigen values $\lambda \rightarrow |A - \lambda I| = 0$

$$\therefore \begin{vmatrix} 1-\lambda & 1 & 1 & 1 \\ 1 & 1-\lambda & 1 & 1 \\ 1 & 1 & 1-\lambda & 1 \\ 1 & 1 & 1 & 1-\lambda \end{vmatrix} \Rightarrow (1-\lambda) \begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} + 1 \begin{vmatrix} 1 & 1-\lambda & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix}$$

$$- 1 \begin{vmatrix} 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix}$$

$$\Downarrow \begin{matrix} R_2 - R_1 \\ R_3 - R_1 \\ R_4 - R_1 \end{matrix}$$

$$\begin{vmatrix} 1-\lambda & 1 & 1 & 1 \\ \lambda & -\lambda & 0 & 0 \\ \lambda & 0 & -\lambda & 0 \\ \lambda & 0 & 0 & -\lambda \end{vmatrix} \Rightarrow (1-\lambda)(-\lambda^3) + (-\lambda^4) + (-\lambda^3) + (-\lambda^3)$$

$$\therefore \lambda^4 - 4\lambda^3 \Rightarrow \lambda^3(\lambda - 4)$$

$$\therefore \boxed{\text{Eigen Values} \rightarrow 0, 0, 0 \& 4}$$

$$C \Rightarrow \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \rightarrow \text{Echlon form} \xrightarrow[R_4 - R_2]{R_3 - R_1} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore \underline{\text{Rank} = 2}$$

Eigen Values $\rightarrow |A - \lambda I| = 0$

$$\therefore \begin{vmatrix} 1-\lambda & 0 & 1 & 0 \\ 0 & 1-\lambda & 0 & 1 \\ 0 & 0 & 1-\lambda & 0 \\ 0 & 1 & 0 & 1-\lambda \end{vmatrix} \Rightarrow (1-\lambda) \begin{vmatrix} 1-\lambda & 0 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} + 1 \begin{vmatrix} 0 & 1-\lambda & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1-\lambda \end{vmatrix}$$

$$\Rightarrow (1-\lambda) [(1-\lambda)((1-\lambda)^2 - 1)] + 1 [-1((1-\lambda)^2 - 1)]$$

$$\Rightarrow (1-\lambda)^2 [(1-\lambda)^2 - 1] - [(1-\lambda)^2 - 1]$$

$$[(1-\lambda)^2 - 1] [(1-\lambda)^2 - 1] \Rightarrow 0$$

$$(\lambda^2 - 2\lambda)^2 \Rightarrow 0$$

$$\therefore \lambda^2 (\lambda - 2)^2 \Rightarrow 0$$

$$\therefore \boxed{\text{Eigen Values} \rightarrow 0, 0, 2, 2}$$